ALESSANDRO CASTELLANI

Milan / Como, Italy · +39 333 528 8084 · alecaste041202@gmail.com

Portfolio: apmcastellani-projects.github.io · **LinkedIn**: in/alessandro-castellani-4905a7246 · **GitHub**: apmcastellani-projects

EDUCATION

Università Cattolica del Sacro Cuore — *Milan, Italy*

Master of Science (Laurea Magistrale) in Data Analytics for Business — *Sep 2025 – Expected 2027*

Coursework: Applied Linear Models, Algorithms & Deep Learning, Time Series, Computational Statistics, Optimization, Statistical Learning.

Università degli Studi dell'Insubria — *Como, Italy*

Bachelor of Science (Laurea Triennale) in Mathematics — *Graduated – Mar 2025*

Thesis: *Neural Network Architectures for Football Performance Forecasting using TensorFlow and Keras.* Modeled multi-season historical player performance metrics to forecast seasonal outputs using backpropagation and Adam optimization.

Pontificio Collegio Gallio — *Como, Italy*

Liceo Scientifico Internazionale (High School Diploma in Science) — *Sep 2016 – Jun 2020*

FEATURED PROJECTS & DATA COMPETITIONS

Evaluating Offensive Production in Serie A (xG Linear Modeling) — *Academic Research (Cattolica) 2026*

Technologies: R, FBref/Opta Event Data, Multiple OLS Regression, Residual Diagnostics

- Built a multiple OLS regression model on 120 Serie A forwards predicting Expected Goals (xG) beyond raw shot volume ($R^2 = 91.4\%$, Adj $R^2 = 90.8\%$).
- Performed tactical outlier diagnostics (identifying tactical roles of Krunic, Cutrone, Joao Pedro) and validated model stability out-of-sample on Olivier Giroud.

IFCS 2026 Data Challenge — *Competitive Machine Learning (Milan-Bicocca) 2026*

Technologies: Python, Scikit-Learn, Imbalanced Classification, Pandas

- Engineered an end-to-end ML classification pipeline predicting financial distress across 14,000 Italian SMEs under time constraints, pitching findings to an academic jury.
- Implemented non-linear signed logarithmic transformations ($\text{sign}(x) \cdot \ln(1 + |x|)$) for heavy-tailed balance sheets, 5-Fold Stratified CV, and Youden's J threshold optimization.

Multi-Horizon XGBoost Predictive Engine & Dashboard — *Time-Series ML Engineering 2026*

Technologies: Python, FastAPI, XGBoost, Purged Walk-Forward Cross-Validation, Plotly

- Built an end-to-end time-series return predictor utilizing purged walk-forward cross-validation with embargo buffers to eliminate lookahead bias; deployed REST API with interactive dashboard.

WORK EXPERIENCE

I Bravi SRL / SC Management SRL — *Various Locations, Italy*

Event & Festival Hospitality Staff — *Apr 2025 – Present*

- Delivered high-volume beverage service and guest hospitality at major music festivals (Nameless Festival, Snowland Festival; 20,000+ daily attendees), handling cash operations and POS reconciliation.

XROAD SSDRL — *Cadorago, Italy*

Operations & Front-Office Assistant (Part-Time) — *Jan 2025 – Feb 2026*

- Managed customer onboarding, CRM database records, and active subscription coordination for over 200 members.

Private Mathematics Tutoring — *Milan / Como, Italy*

Mathematics Tutor — *Sep 2020 – Present*

- Mentored 50+ secondary and undergraduate university students in advanced calculus, linear algebra, and probability theory.

ATHLETICS & EXTRACURRICULAR

- **Hyrox Competitor**: Functional fitness racing combining 8km running and 8 high-intensity workout stations.

TECHNICAL SKILLS & COMPETENCIES

- **Programming & Tools**: Python (NumPy, Pandas, Scikit-Learn, TensorFlow, Keras, FastAPI), R (tidyverse, car, leaps), SQL, Git, LaTeX.
- **Domain Expertise**: Football Analytics (Opta/FBref event data, xG/xA modeling, role benchmarking), Financial Modeling, Time Series.
- **Languages**: Italian (Native), English (Fluent / Professional Working Proficiency).